Casino Gaming Analysis Report

Andrea Lucini Paioni

Dataset Overview and Scope

This report analyzes one month of aggregated casino activity data, focusing on four European and South American markets (Sweden, Estonia, Finland, and Chile). The goal is to establish data integrity, define key performance indicators, and extract actionable business insights.

The raw dataset contained 13 variables in 720 rows, each for every hour in September 2025, with categorical fields like *provider*, *game_type*, and *market_code*, and numerical fields like *bets_eur*, *wins_eur*, and *bet_count*.

Data Cleaning

To ensure the integrity of the analysis, several corrections were made before starting the development of the data visualization:

1. The numerical variable *avg_bet_size* was found to be inconsistent (*bets_eur / bet_count*), so it was corrected by recalculating the value for all the records.
2. The categorical variable *casino_game_name* was standardized, normalizing all four variations of *'Starburst'* to a single value and cleaning to obtain names with uppercase initials and lowercase characters for the rest of the names.
3. The relationship between *provider* and *provider_id* was sometimes incorrect, *provider* was considered the correct one due to consistency with other variables (like *game_type*).
4. The *customer_dim_id* field, which seems to be an hourly index that stopped mid-dataset, was rebuilt to ensure every record has a unique, ordered identifier.
5. The *market* fields were enriched with the information of the full country name.

KPI developed

The analysis focuses on four financial KPI:

- **Total bets:** the total number of bets made by customers, considered as primary metric for measuring player engagement
- **Total money bet:** The aggregate value (€) of all money wagered by customers, also considered as on of the primary metrics for measuring player engagement.
- **Total win:** Calculated as the total amount of money (€) gained by the customers. This should be higher than the total money bet KPI to be considered profitable.
- **Margin (and Margin %):** Calculated as $\frac{(Total\ money\ bet - total\ money\ won))}{Total\ money\ bet}$ (€ for the absolute value, % for the second one). This is the core metric used to assess the casino's profitability.

Other KPI were developed, like **Money gained** (as *Total money bet - Total money won*) and **Average Money gained**, but the four above can be considered as the most important and significant.

Main Analysis and Insights

The Swedish market stands out as a significant outlier, showing behaviors drastically different from the other three markets (Estonia, Finland, and Chile). Without Sweden, the average total money bet across the remaining markets is approximately **€12,000**, with a total of **€8,500** won per country. At the same time Swedish customers spent nearly **€15,500** but won significantly more (**€18,500**).

As a result, Estonia, Finland and Chile are highly profitable markets, registering a consistent **20% Margin** throughout the month, with almost zero days of net losses. Swedish market was highly **unprofitable** in September, resulting in a net negative margin for the company in every game type except Slots.

A clear difference exists between the two main categories of games, based on bet behavior:

- **Slots games (Play'n GO, NetEnt...)** accounts for the vast majority of total bets placed, but with low value, as the average bet size is approximately **€4** per wager; this type of games is highly lucrative for the company, with customers losing an average of €3 for every €4 bet placed. Play'n GO is the primary example of provider for this kind of games.
- **Live & Table Games (Evolution provider):** this type of games hold a much higher average bet size, ranging from **€50 to €90** per wager, but at the same time they particularly impact the negative result of margin observed in Sweden. Evolution is the primary example of provider for this kind of games (like Poker, Crazy Time and Blackjack).

Another aspect to consider is the distribution of bets during the month: the **first third of the month** shows the highest overall money wagered by customers, and also the only time when Swedish customers experienced a net loss, while the rest of the month showed positive outcomes (more won than bet) for Swedish players.

Considering instead **weekly trends, the other markets** had bet peaks on **Mondays** and decreases mid-week, while Sweden was again different from the other markets, with the highest average betting activity occurring on **Thursdays**, with Monday as the second-highest peak.

Strategic Conclusion

The analysis reveals **two main types of markets**: a **stable, high margin market** in **Estonia, Finland, and Chile**, with high-volume and low-stakes slot play mainly; and a **high-stakes, high-volatile**, and currently **loss-making market** in **Sweden**, based mainly on live games like Poker, Blackjack and Crazy Time.

Future analysis should focus on **Swedish customer** to understand the **behavior** of the customers and what drives to these excessive payouts, maybe adjusting bonus structures or game limits; for the **other markets**, the focus should be to introduce some customers, both old and new, to the **live games**, while also **keeping the high-profit** of the **slots** market.